# Sense-Plan-Act Pipeline for the IASYS Roadwork Challenge

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## 1 Introduction

In the IASYS Roadwork Challenge, a vehicle in a 2D ROS2 Jazzy simulator starts at (5, -1.75) m. It must obey a STOP/GO panel before the entry line x = 25m, and pass two static work discs (x ≈ 43-47 m) in the opposing lane. It must then return to its own lane before x = 68 m and stop at the goal (x ≥ 80 m, speed < 0.15 m/s).

This delivery covers **S1** (GO throughout) and **S2** (STOP for t < 10 s, then GO; no crossing of x = 25m during STOP).

## 2 System architecture

The solution is three independent C++ executables in `roadwork_student`, connected only by ROS topics (Fig. 1).

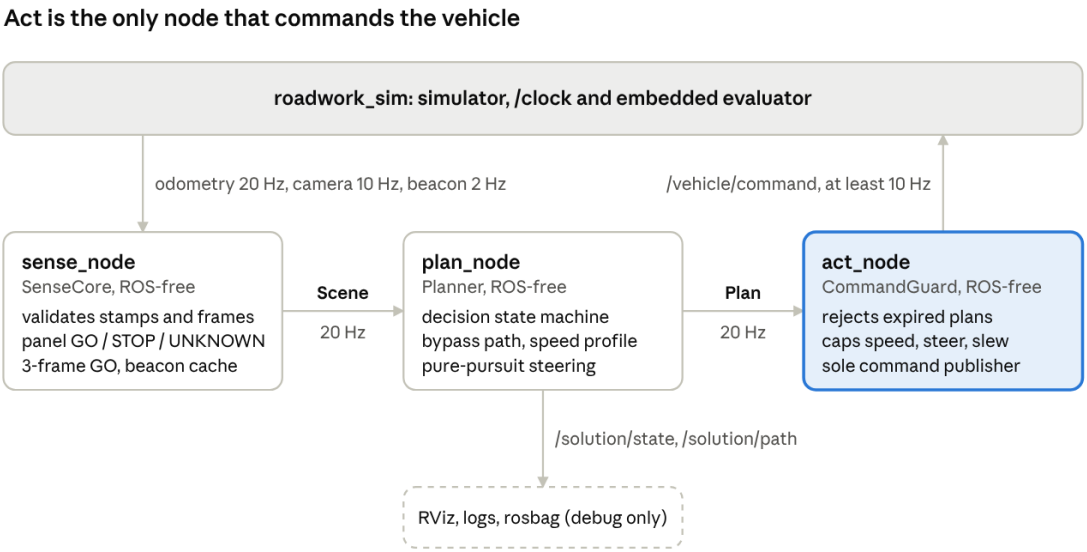



Fig. 1 · system architecture

Each node wraps a ROS-free core (SenseCore, Planner, CommandGuard in include/roadwork_student/), so the same code is unit- and closed-loop tested without ROS. The chain is event-driven at 20 Hz (odometry > scene > plan > command). A 50 ms timer in each node publishes a safe output (invalid scene, hold plan, braking command) when its input stops.

# 3 Interfaces

Every boundary carries source stamps and an explicit expiry in simulation time. Sense subscribes to odometry and camera with SensorDataQoS, since a reliable-only subscriber would not match these best-effort publishers; the beacon is reliable with depth 10. LiDAR and radar are unused in D1. The internal topics `/solution/scene` and `/solution/plan` are reliable with depth 1, because only the newest message matters.

The student package `roadwork_student_interfaces` defines two messages:

- **Scene** (sense > plan) carries the ego state with `odom_stamp` and validity, the raw and filtered panel class (UNKNOWN/GO/STOP) with `camera_stamp`, and the cached beacon geometry with `beacon_stamp`. Each block keeps its original sensor stamp.
- **Plan** (plan > act) carries the targets (speed, steering), a speed cap, a hold flag, the state and reason, the stamps used and `valid_until` = stamp + 0.25 s.

| Check | Node | Threshold | Action |
|---|---|---|---|
| Odometry age | Sense, Plan | 0.20 s | SAFE_STOP |
| Camera age | Sense | 0.30 s | panel > UNKNOWN |
| GO frame gap | Sense | 0.15 s | GO count restarts |
| Future stamp | all | now + 0.06 s (one clock tick + slack) | sample rejected |
| Scene age | Plan | 0.25 s | hold plan |
| Plan expiry / age | Act | `valid_until`, 0.30 s | speed 0, steering held |

# 4 Design decisions

The guiding rule is asymmetric: evidence that permits motion (GO) must be confirmed and fresh, while evidence that forbids it (STOP, missing or stale data) acts at once.

## 4.1 Perception

The panel is a synthetic 96×64 crop with a centered disc and ±18 noise, so a hand-written classifier suffices. Inside a 14 px disc, a pixel counts as red or green if its dominant channel is ≥ 110 and ≥ 70 above both others. A frame ≥ 80 % green is GO, ≥ 80 % red is STOP, otherwise UNKNOWN. Frames with >10 % color in the outer ring are classified UNKNOWN. STOP acts at once, GO needs three consecutive GO frames ≤ 0.15 s apart. Valid beacons are cached, so the path follows work_shift without scenario ids.

## 4.2 Decisions

Without a confirmed GO (STOP, UNKNOWN or stale), Plan's state machine (Fig. 2) stays in WAIT_GO: the vehicle drives in its own lane and a 1.0 m/s² stop-line profile brings it to rest with its front 1 m before the entry line. Permission is re-checked every cycle, so losing GO before commitment returns Plan to WAIT_GO. GO is latched, and the path frozen, only once the front is within braking distance of the line

(0.15 s control delay plus 3 m/s² braking). Up to that point a STOP is still obeyed, and after it a noisy frame cannot trigger re-planning.

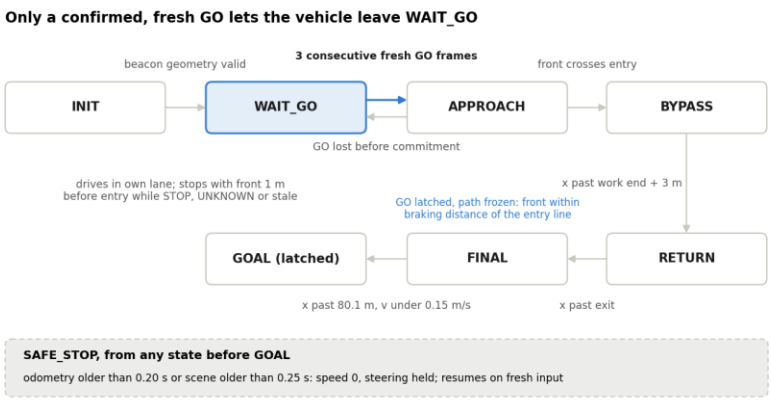



Fig. 2 · Plan decision state machine

### 4.3 Motion and command guard

The path leaves the lane (y = -1.75 m) at entry -0.5m, holds the bypass (y = +1.25 m) from work start -2m to work end +3m and returns by exit -2.5m. The smootherstep $p(s) = 10s^3 - 15s^4 + 6s^5$ keeps curvature continuous (peak steering 0.215 rad). Speed is the minimum of 3.5 m/s cruise, a curvature limit 15 m ahead, 1.0 m/s² goal braking, the stop-line profile while in WAIT_GO and the beacon limit -0.3 m/s; steering is pure pursuit with a 3-6 m look-ahead. Act brakes on any missing, non-finite, future or expired plan and caps speed (3.8 m/s), steering (±0.45 rad) and steering rate (0.6 rad/s).

## 5 Experimental setup and results

Each scenario was run in ROS 2 Jazzy with seeds 1-3, one fresh launch per episode and visualization off. The table gives the evaluator metrics from each result.json and the GO latch logged by Plan; all files are in results/d1.

| Run | Result | Time (s) | Min clearance (m) | Max speed (m/s) | Stopped (s) | GO latched (t, x) |
|---|---|---|---|---|---|---|
| S1 seed 1 | PASS | 25.05 | 1.167 | 3.50 | 0.60 | 6.45 s, 21.65 m |
| S1 seed 2 | PASS | 25.05 | 1.167 | 3.50 | 0.60 | 6.45 s, 21.65 m |
| S1 seed 3 | PASS | 25.00 | 1.167 | 3.50 | 0.55 | 6.40 s, 21.65 m |
| S2 seed 1 | PASS | 29.55 | 1.168 | 3.50 | 2.05 | 11.05 s, 23.69 m |
| S2 seed 2 | PASS | 29.55 | 1.168 | 3.50 | 2.05 | 11.05 s, 23.69 m |
| S2 seed 3 | PASS | 29.55 | 1.168 | 3.50 | 1.90 | 11.05 s, 23.70 m |

All six episodes passed with no collision, off-road, entry, lane, speed or timeout violation. In S1, Plan confirmed GO at 0.45-0.55 s, once the first 2 Hz beacon had arrived; the 0.55-0.60 s of stopped time is this start-up wait. In S2, Plan entered WAIT_GO at 0.55-1.10 s and drove to the stop line. The panel turned GO (third GO frame) at 10.15-10.20 s, and Plan left WAIT_GO at 10.20 s with the vehicle at x =

23.12 m (front 24.02 m, 0.98 m before the line). It latched GO at 11.05 s, crossed the entry line at 11.35 s and then reached each later state about 4.5 s after S1 (Fig. 3). Rejected commands are start-up ticks before the first valid plan.

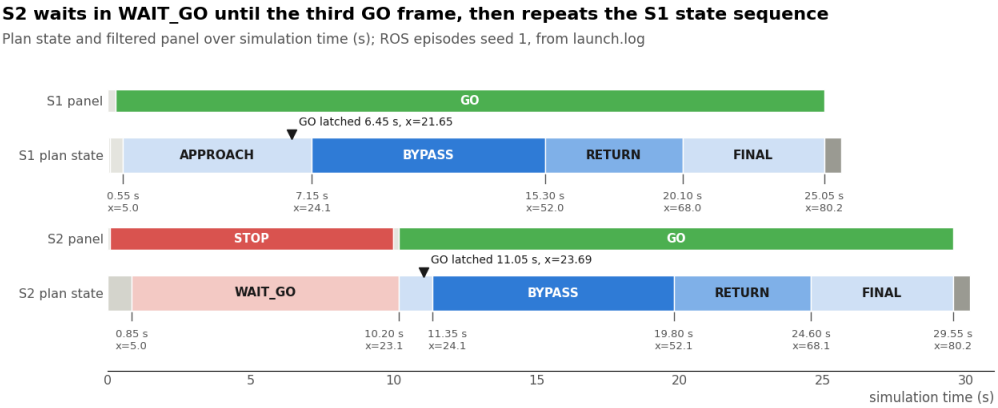



Fig. 3 · Plan state and panel timeline, S1 and S2 seed 1

## 6 Discussion and limitations

The pipeline meets S1-S2 in every tested condition, but two gaps remain.

- **Commitment rules.** Entry requires fresh camera evidence but accepts a cached beacon of any age, and once GO is latched (within braking distance of the line) a later STOP is ignored. Neither occurs in S1 and S2, but S4 needs a beacon younger than 1.2 s at entry.
- **Start-up wait.** Until the first beacon arrives the vehicle stays in INIT or SAFE_STOP with Act braking. This is safe, but costs about 0.5 s and explains the rejected commands.

For D2, Sense will add LiDAR and radar. Plan will add a beacon-freshness check at commitment, a radar gap check (S3), evidence-loss recovery (S4) and obstacle braking (S5).

## 7 Conclusion

The pipeline completes S1 and S2 without violations in all six ROS episodes. For D2 it will add a radar gap check (S3), evidence-loss recovery (S4) and obstacle braking (S5).